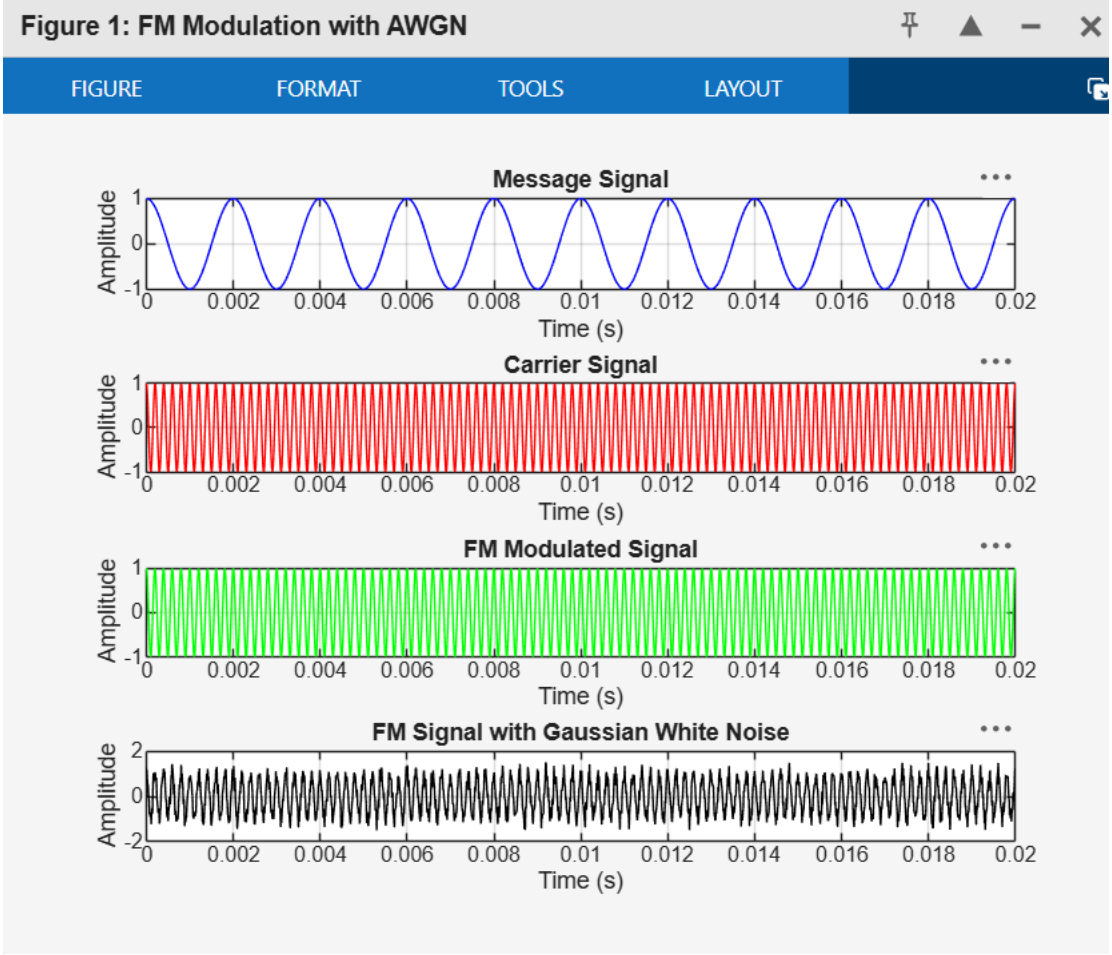
EXP 10 ADC


Figure 1: FM Modulation with AWGN
FIGURE
FORMAT
TOOLS
LAYOUT
Message Signal
Carrier Signal
FM Modulated Signal
FM Signal with Gaussian White Noise
Amplitude
Time (s)
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018
0.02

/MATLAB Drive/ADCexp10.m

```
clc;
clear;
close all;

% 1. PARAMETERS
fs = 100000;              % Sampling frequency (Hz)
t = 0:1/fs:0.02;          % Time vector
fm = 500;                 % Message frequency (Hz)
fc = 5000;                % Carrier frequency (Hz)
kf = 100;                 % Frequency sensitivity (Hz/V)
SNR = 10;                 % Signal-to-noise ratio (dB)

% 2. MESSAGE SIGNAL
m = cos(2*pi*fm*t);

% 3. CARRIER SIGNAL
c = cos(2*pi*fc*t);

% 4. FM MODULATION
% Integrate the message signal to obtain its phase contribution
phase = 2*pi*kf*cumsum(m)/fs;
fm_sig = cos(2*pi*fc*t + phase);

% 5. ADD AWGN NOISE
signalPower = mean(fm_sig.^2);
noisePower = signalPower / (10^(SNR/10));
noise = sqrt(noisePower)*randn(size(fm_sig));
noisy = fm_sig + noise;

% 6. DISPLAY RESULTS
figure('Name','FM Modulation with AWGN');

subplot(4,1,1);
plot(t,m,'b');
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;

subplot(4,1,2);
plot(t,c,'r');
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
```

```
47
48  subplot(4,1,3);
49  plot(t,fm_sig,'g');
50  title('FM Modulated Signal');
51  xlabel('Time (s)');
52  ylabel('Amplitude');
53  grid on;
54
55  subplot(4,1,4);
56  plot(t,noisy,'k');
57  title('FM Signal with Gaussian White Noise');
58  xlabel('Time (s)');
59  ylabel('Amplitude');
60  grid on;
61
62  disp('FM modulation with AWGN completed successfully.');
63
```